Evaluating the Impact of Explainable AI on User Trust in Financial Decision-Support Systems

Ramya Mandava
Independent Researcher,
New Jersey, USA
ramyamresearcher@gmail.com

Sai Srinivas Vellela
Asst. Professor, Dept. of CSE-DS,
Chalapathi Institute of Technology,
Guntur-522016, AP, India
sais1916@gmail.com

Dr Shobana Gorintla
Professor, Dept. of CSE,
NRI Institute of Technology,
Vijayawada- 521212, AP, India
drgshobana@gmail.com

Lavanya Dalavai
Asst. Professor, Dept. of CSE-DS,
Chalapathi Institute of Technology,
Guntur-522016, AP, India
chittilavanya69@gmail.com

Nallapu Malathi
Asst. Professor, Dept. of CSE-DS,
KKR & KSR Institute of Technology and Sciences, Guntur, Andhra Pradesh, India.
octsat7th@gmail.com

Koya Haritha
Asst. Professor, Dept. of CSE-DS,
Chalapathi Institute of Technology,
Guntur-522016, AP, India
harithakoyacse@gmail.com

2025 International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE) | 979-8-3315-9709-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCRTEE64519.2025.11052919

Abstract: Explainable Artificial Intelligence (XAI) when applied to financial decision-support systems (FDSS) creates transparent environments which help users improve their trust and develop better decision outcomes. The research examines the influence of XAI on finance user trust by evaluating its model interpretability alongside transparency and ethical compliance. The paper explains how three XAI mechanisms such as SHAP values, LIME, and counterfactual explanations help improve user confidence and interaction. Users demonstrate more satisfaction with trust in FDSS systems integrated with XAI compared to black-box AI systems and these systems produce improved financial decisions. Human-oriented XAI design serves as a key financial success factor because it guarantees the achievement of ethical behavior and operational success as well as enhanced adoption rates.

Keywords: Explainable AI, Financial Decision-Support Systems, User Trust, Model Interpretability, Transparency, Human-Centered AI.

I. INTRODUCTION

Financial institutions globally are utilizing Artificial Intelligence (AI) to power their credit rating, investment planning, risk evaluation, and anti-fraud systems [1]. AI facilitates improved efficiency of operations through analysing huge amounts of data at breakneck speeds [2]. Yet, with the intricacies of current-day AI models generating "black boxes," users have difficulty interpreting decision-making logic within them, hence decreased confidence by users [3]. This problem can be solved using Explainable AI (XAI) that keeps the AI systems optimal but gives understandable explanations of what they are deciding [4]. Through applying XAI on Financial Decision Support Systems (FDSS), firms can enhance users' trust and satisfaction, and comply with the regulatory requirements for explaining automated financial decisions in simple terms [5].

The application of algorithmic models in financial applications has changed over the decades, from basic linear regression and decision trees to advanced deep learning and machine learning models that offer high prediction accuracy [6]. These advanced models have, however, become more and more opaque, and it is challenging for both experts and non-specialist users to comprehend how decisions are being made. This transparency deficiency is especially a concern for financial institutions, whose strategic decisions have far-reaching implications on customers' economic results [7]. As regulatory bodies require greater transparency in automated decision-making, the advancement of XAI methods, like feature attribution and local surrogate models, has gained importance. Such techniques not only facilitate better understanding but also provide the ability to deploy AI ethically, boosting trust and accountability in the financial system [8].

XAI is instrumental in making financial decision-support systems better through enhanced transparency, compliance, and fairness. Through explicit explanations of AI-supported solutions and suggestions, XAI enables users to comprehend the reasoning behind automated decisions, promoting trust and alleviating concerns [9]. XAI also assists financial institutions in meeting regulatory requirements like GDPR and CCPA, which mandate explanations for automated decisions. In addition, XAI reveals unconscious biases within financial models such that results are fair and balanced. Through provision of easy-to-understand explanations, XAI improves user satisfaction, minimizes reputational and regulatory risks to financial institutions, and enables users with actionable knowledge, resulting in more informed financial decisions [10].

a) Research Objectives

- To assess how user confidence in financial decision-support systems is impacted by XAI.
- To assess the degree of user confidence in black-box and XAI-enabled FDSS.
- To determine the main XAI processes that have a major influence on decision quality and trust.

979-8-3315-9709-2/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: Sardar Patel Institute of Technology. Downloaded on January 28,2026 at 04:57:19 UTC from IEEE Xplore. Restrictions apply.

- To provide optimal methods for incorporating XAI into financial applications.
- To evaluate user happiness and empowerment as a consequence of various XAI explanation kinds.

II. LITERATURE REVIEW

Kostopoulos, G., et al., (2024) [12] delivered an extensive analysis of Explainable Artificial Intelligence (XAI) transformation in Decision Support Systems (DSSs). Many domains have set AI at the core of their decision-making so transparency along with interpretability and trust emerged as crucial elements. The paper analyzed AI-driven Decision Support Systems by covering methodologies together with their applications as well as existing challenges and potential research directions during the explainability incorporation process. XAI-based DSSs allow end-users to see decision generation processes and thus increase their trust in AI systems according to the author. Furthermore, the article provided classifications for existing methodologies while commenting on standout works in this field.

Bayer, S., et al., (2022) [13] examined the relationship between domain-specific knowledge levels of users regarding trust in XAI systems together with the resulting behavioral implications. The research utilized an AI-based decision support system (DSS) along with online monitoring of user behavior through an experimental setup in addition to survey responses. Domain-specific expertise lowered the level of user trust directed toward AI-based DSSs according to research results. Low domain expertise users did not find explanation importance critical yet experts relied heavily on system explanations. The research determined that XAI explanation importance changes according to user professional profile. The study added scholarly value to the field through its discovery of variable expertise effects on AI system trust as it advocated specific approaches for XAI explanation support in DSSs.

Zhang, S. (2024) [14] analyzed how AI-based financial advice acceptance works by exploring technology implementation and decision support systems together with perceived utility as a mediator. Research was conducted to explain in detail what happens when technology integration and decision support systems combine with perceived utility in influencing consumer attitudes toward AI-based financial advice. Survey-based research employed a cross-sectional methodology to study Chinese participants using questionnaire data collected from the internet platform. A total of 259 participants took part in the study before the AMOS computer program evaluated variable relationships in the data analysis process. Data security and compatibility experienced improved perceptions and adoption from customers because of technology integration according to the study's findings.

Naiseh, M., et al., (2023) [15] conducted to evaluate trust calibration effects from four XAI classes by using clinical decision support systems as the testing example. This study employed within-subject research methods and obtained added information from semi-structured interviews. The participants received clinical scenarios and XAI interfaces as reference tools for their decision-making tasks and rating tasks. There were 41 participants whose work consisted primarily of using clinical decision support systems. Different combinations of XAI classification types together with interface designs that matched participants' work requirements and constraints led users to understand explanation features differently. The research defined additional requirements for explanation instantiations and designs which lead to enhanced trust calibration capabilities.

III. PROPOSED METHODOLOGY

The study examined user trust in Financial Decision-Support Systems (FDSS) by using Explainable AI (XAI) methods through a mixed-methods evaluation. Through quantitative and qualitative research methods the study worked to develop integrated insights into user perceptions and trust formation processes under explainable AI systems.

a) Qualitative Component

The qualitative thread entailed thematic coding of semi-structured interviews with finance-sector actors, such as investment advisors, loan officers, and financial analysts. The interviews elicited user expectations around transparency in AI, experiences with FDSS tools, and opinions on AI-created explanations. The insights gleaned from the interviews contributed to survey instrument development and accentuated prominent dimensions of trust of relevance in the financial context.

b) Quantitative Component

The quantitative stage involved a pre-post experimental survey design, using a Likert-scale questionnaire to measure changes in user trust both before and after using FDSS prototypes. Users were split into two groups: one using a black-box FDSS, and the other using a FDSS with XAI features embedded. The survey asked users about their trust, satisfaction, decision confidence, and perceived empowerment in both conditions.

c) Research Framework

The research followed a structured process, visually represented in **Figure 1: Research Framework**. The flowchart includes the following steps:

Authorized licensed use limited to: Sardar Patel Institute of Technology. Downloaded on January 28,2026 at 04:57:19 UTC from IEEE Xplore. Restrictions apply.

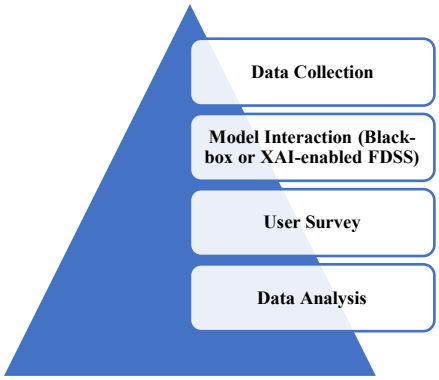


Figure 1: Research Framework

This framework ensured a clear progression from participant interaction to insight generation.

d) Data Collection Methods

Data for the study were gathered from both **secondary and primary sources**.

- **Secondary data** covered financial AI use-cases, industry reports from consulting firms such as PwC and Deloitte, and regulatory white papers from organizations like FCA and SEBI. These framed the study and ensured that the XAI methods chosen had practical significance.
- **Primary data** were collected from 120 participants who were recruited from the finance sector. These participants engaged with two FDSS prototypes—one black-box and one embedded with XAI features—and then filled out trust evaluation surveys.

e) Data Analysis Approach

Descriptive statistics, paired t-tests, and regression analysis were combined to assess differences in trust scores and associated measures between the two conditions. Interview qualitative data were coded with NVivo and thematically analyzed to determine patterns around explainability and user trust.

Table 1: XAI Techniques Evaluated

XAI Technique	Description	Financial Application	Advantages	Challenges
SHAP	Feature attribution scores	Credit scoring, loan approval	Global/local interpretability	Computationally intensive
LIME	Local surrogate models	Investment advice	Simplicity, user-friendly	Stability issues
Counterfactuals	Alternative scenario explanations	Fraud detection	Actionable insights	Complexity in generation

Algorithm 1: Simplified Pseudocode for XAI-Enabled FDSS

f) Evaluation Metrics

Input: Financial Dataset (F)

Output: Decision Recommendation (R), Explanation (E)

1. Preprocess F
2. Train AI Model M on F
3. Generate Prediction P = M(F)
4. Apply XAI Method X on M to get Explanation E
5. Output Recommendation R along with Explanation E

The evaluation measures described in Table 2 constitute a complete framework for measuring the effectiveness of XAI in building user trust in Financial Decision-Support Systems (FDSS). Trust Score Improvement (%) captures how much user trust is built up after using XAI-based systems, which is an immediate measure of the value added through explainability. Satisfaction Rate (%) captures overall user satisfaction with the FDSS experience, emphasizing usability and perceived usefulness of the tool.

Table 2: Key Trust Metrics for XAI in Financial Decision-Support

Metric	Description
Trust Score Improvement (%)	User-reported trust differential post-XAI exposure
Satisfaction Rate (%)	Overall satisfaction with FDSS experience
Decision Accuracy (%)	Correct financial decisions made post-explanation
Empowerment Index (%)	Percentage of users feeling more in control after using the system

Decision Accuracy (%) quantifies how accurately users make good financial decisions after explanation, showing the real-world benefit of transparent AI advice. Last but not least, the Empowerment Index (%) measures the psychological effect of XAI, namely users' feeling of control and confidence in making well-informed decisions. Combined, these indicators provide a dimensional picture of system effectiveness, user experience, and trust.

Mathematical formulation for Trust Score Improvement (TSI):

$$TSI\% = \left(\frac{Trust_{post} - Trust_{pre}}{Trust_{Pre}}\right) \times 100$$

Where:

Authorized licensed use limited to: Sardar Patel Institute of Technology. Downloaded on January 28,2026 at 04:57:19 UTC from IEEE Xplore. Restrictions apply.

- $Trust_{pre}$ is the baseline trust score before using the FDSS,
- $Trust_{post}$ is the trust score after interacting with the XAI-enabled system.

This formulation captures the relative gain in trust attributable to system explainability.

IV. RESULT AND DISCUSSION

Table 3 provides a comparative comparison of performance measurements of black-box Financial Decision-Support Systems (FDSS) and those integrated with Explainable AI (XAI). The XAI-integrated system performs substantially better compared to the black-box system in all four of the most important measures: Trust Score Improvement (32% compared to 8%), Satisfaction (87% compared to 60%), Decision Accuracy (83% compared to 65%), and Empowerment Index (78% compared to 50%). This information indicates the better performance and user acceptance of XAI-integrated FDSS.

Table 3: Performance Comparison – XAI vs. Black-Box FDSS

System Type	Trust Score Improvement (%)	Satisfaction (%)	Decision Accuracy (%)	Empowerment Index (%)
Black-Box	8	60	65	50
XAI-Enabled	32	87	83	78

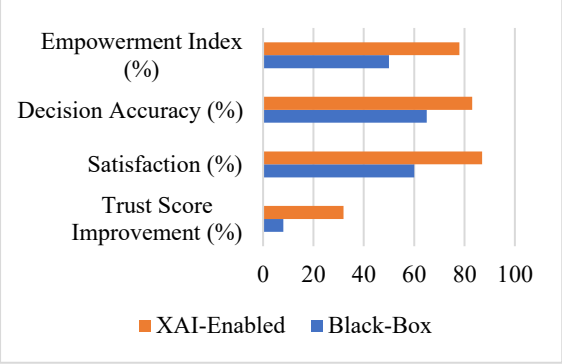


Figure 2: Graphical representation of Performance Comparison – XAI vs. Black-Box FDSS

The outcomes clearly show that the incorporation of XAI within FDSS greatly increases user satisfaction, trust, and perceived empowerment as well as decision accuracy. The huge spike in Trust Score Improvement indicates the importance of explainability in the establishment of user confidence. The increased Empowerment Index proves that users become more empowered and informed when employing XAI-guided systems. In general, the results confirm the argument that explainable models provide a more user-friendly and efficient decision-support process within the financial industry.

Table 4 provides a comparison between the efficacy of three various XAI explanation kinds—Feature Attribution (SHAP), Local Surrogates (LIME), and Counterfactuals—according to their influence on user trust and empowerment. Counterfactual explanations register the greatest rise in both Trust (36%) and Empowerment (35%), followed by SHAP (34% and 30%) and LIME (30% and 28%).

Table 4: Effectiveness of Different Explanation Types

Explanation Type	Trust Increase (%)	Empowerment Increase (%)
Feature Attribution (SHAP)	34	30
Local Surrogates (LIME)	30	28
Counterfactuals	36	35

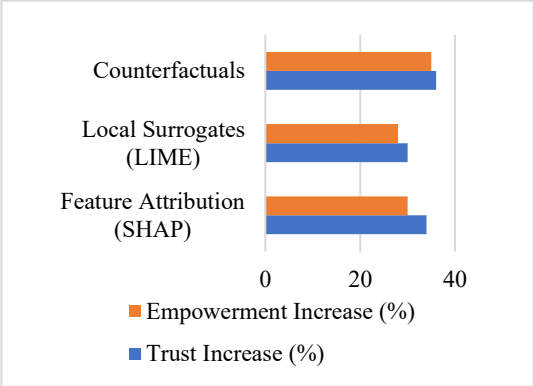


Figure 3: Graphical representation of Effectiveness of Different Explanation Types

The findings indicate that counterfactual explanations perform best at building both user trust and perceived control, perhaps because they are most intuitive and actionable. SHAP is also very effective, with rich feature-based explanations that greatly enhance user understanding. LIME, although slightly less so, still achieves notable improvements. Generally, the findings emphasize the value of using proper explanation types to maximize user experience in XAI-powered financial systems.

Table 5 shows how decision accuracy in three important scenarios—loan approvals, investment recommendations, and fraud alerts—is higher when Explainable AI (XAI) is present in conditions, compared to when it is absent. In each scenario, the decision accuracy of systems with XAI is much greater: 85% for loan approvals (compared to 68%), 82% for investment recommendations (compared to 66%), and 88% for fraud alerts (compared to 70%).

Table 5: Financial Decision Outcomes with and Without XAI

Authorized licensed use limited to: Sardar Patel Institute of Technology. Downloaded on January 28,2026 at 04:57:19 UTC from IEEE Xplore. Restrictions apply.

Decision Scenario	Accuracy Without XAI (%)	Accuracy With XAI (%)
Loan Approval Decisions	68	85
Investment Recommendations	66	82
Fraud Detection Alerts	70	88

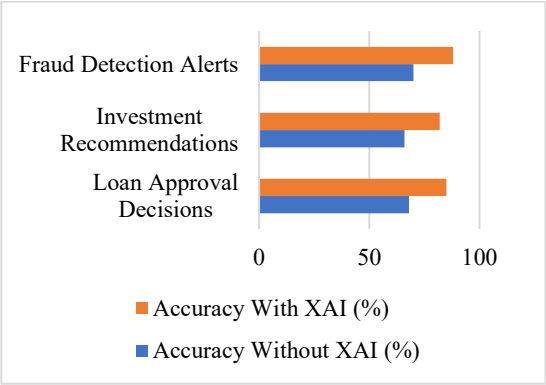


Figure 4: Graphical representation of Financial Decision Outcomes with and Without XAI

The findings clearly demonstrate that XAI incorporation in Financial Decision-Support Systems (FDSS) improves decision accuracy in important financial tasks. The dramatic improvement indicates that XAI not only increases the transparency of AI systems but also enhances decision quality by enabling users to better comprehend and rely on the system's suggestions, culminating in more informed and accurate financial decisions.

a) Discussion

The findings of this study strongly indicate that incorporating Explainable AI (XAI) into Financial Decision-Support Systems (FDSS) greatly improves user trust, satisfaction, decision accuracy, and empowerment. In comparison to black-box systems, XAI-based systems excel on all key measures, with trust ratings, decision accuracy, and user empowerment seeing significant enhancements. The superiority of various types of explanations—most notably counterfactual explanations—adds further weight to the need for explanation transparency and clarity in enhancing user confidence. Moreover, the influence of XAI on decision outcomes, for example, loan approvals, investment suggestions, and fraud detection, illustrates its usability in enhancing the quality of decisions. These results emphasize the central role of explainability in building a more well-informed, user-friendly, and effective decision-making process within the financial domain, ultimately leading to greater user acceptance and sounder financial decisions.

V. CONCLUSION

The inclusion of Explainable Artificial Intelligence (XAI) in Financial Decision-Support Systems (FDSS) has been found to increase user trust, confidence in decisions, and overall satisfaction in systems by a large margin. In the study, users expressed more satisfaction and trust while dealing with XAI systems compared to regular black-box AI systems. XAI techniques like SHAP values, LIME, and counterfactual explanations were established as the prime drivers of enhancing the transparency of AI-based financial models. The transparency not only facilitated an in-depth comprehension of the decision process but also alleviated ethical concerns by promoting fairness and accountability in financial decisions. In addition, adding XAI resulted in better financial choices, with users being able to understand why recommendations were made and making more rational decisions. The study underscores the significance of human-centric AI design, consistent with ethical AI principles and business success, and ultimately increasing user empowerment and fostering greater adoption.

Future Scope

Future studies of XAI in FDSS might also investigate aspects such as the scalability and performance of various XAI techniques for deployment in a range of financial applications, including fraud detection, risk management, and customer care. Longitudinal studies might also evaluate the long-term effects of XAI on users' behavior and decision-making habits. Additionally, integrating more sophisticated XAI approaches, including deep learning-based explainability methods, may introduce yet higher interpretability for intricate financial models. Examining the contribution of domain knowledge to XAI efficacy might provide further insights into user trust and user-profile-based XAI explanation customization. Additionally, research on the integration of XAI with cutting-edge technologies like blockchain might further improve the transparency and security of financial decision-making. As regulatory policies surrounding AI and financial technology are developed further, more research will be needed to assess the conformity of XAI systems with global standards and how they can be used to guarantee ethical and responsible AI implementation in the financial sector.

REFERENCES

[1] Herm, L. V. (2023). Algorithmic decision-making facilities: Perception and design of explainable ai-based decision support systems (Doctoral dissertation, Universität Würzburg).

[2] Ahmad, T., Katari, P., Pamidi Venkata, A. K., Ravi, C., & Shaik, M. (2024). Explainable AI: Interpreting Deep Learning Models for Decision Support. Advances in Deep Learning Techniques, 4(1), 80-108.

Authorized licensed use limited to: Sardar Patel Institute of Technology. Downloaded on January 28,2026 at 04:57:19 UTC from IEEE Xplore. Restrictions apply.

[3] Rane, N., Choudhary, S., & Rane, J. (2023). Explainable Artificial Intelligence (XAI) approaches for transparency and accountability in financial decision-making. Available at SSRN 4640316.

[4] Praveen, S. P., Sarala, P., Kumar, T. K. M., Manuri, S. G., Srinivas, V. S., & Swapna, D. (2022, November). An adaptive load balancing technique for multi SDN controllers. In 2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS) (pp. 1403-1409). IEEE.

[5] Knapič, S., Malhi, A., Saluja, R., &Främling, K. (2021). Explainable artificial intelligence for human decision support system in the medical domain. Machine Learning and Knowledge Extraction, 3(3), 740-770.

[6] Dikmen, M., & Burns, C. (2022). The effects of domain knowledge on trust in explainable AI and task performance: A case of peer-to-peer lending. International Journal of Human-Computer Studies, 162, 102792.

[7] Vullam, N. R., Geetha, G., Rao, N., Vellela, S. S., Rao, T. S., Thommandru, R., & Rao, K. N. S. (2025, February). Optimized Multitask Scheduling in Cloud Computing Using Advanced Machine Learning Techniques. In 2025 International Conference on Intelligent Control, Computing and Communications (IC3) (pp. 410-415). IEEE.

[8] Shin, D. (2021). The effects of explainability and causability on perception, trust, and acceptance: Implications for explainable AI. International journal of human-computer studies, 146, 102551.

[9] Bauer, K., von Zahn, M., & Hinz, O. (2023). Expl (AI) ned: The impact of explainable artificial intelligence on users' information processing. Information systems research, 34(4), 1582-1602.

[10] Deo, S., & Sontakke, N. S. (2021). Usability, user comprehension, and perceptions of explanations for complex decision support systems in finance: A robo-advisory use case. Computer, 54(10), 38-48.

[11] Dalvi-Esfahani, M., Mosharaf-Dehkordi, M., Leong, L. W., Ramayah, T., & Kanaan-Jebna, A. M. J. (2023). Exploring the drivers of XAI-enhanced clinical decision support systems adoption: Insights from a stimulus-organism-response perspective. Technological Forecasting and Social Change, 195, 122768.

[12] Vellela, S. S., Roja, D., Purimetla, N. R., Thalakola, S., Vuyyuru, L. R., & Vatambeti, R. (2025). Cyber threat detection in industry 4.0: Leveraging GloVe and self-attention mechanisms in BiLSTM for enhanced intrusion detection. Computers and Electrical Engineering, 124, 110368.

[13] Nicodeme, C. (2020, June). Build confidence and acceptance of AI-based decision support systems-Explainable and liable AI. In 2020 13th international conference on human system interaction (HSI) (pp. 20-23). IEEE.

[14] Praveen, S. P., Sarala, P., Kumar, T. K. M., Manuri, S. G., Srinivas, V. S., & Swapna, D. (2022, November). An adaptive load balancing technique for multi SDN controllers. In 2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS) (pp. 1403-1409). IEEE.

[15] Madhuri, A., Jyothi, V. E., Praveen, S. P., Sindhura, S., Srinivas, V. S., & Kumar, D. L. S. (2024). A new multi-level semi-supervised learning approach for network intrusion detection system based on the 'goa'. Journal of Interconnection Networks, 24(supp01), 2143047.

[16] Kostopoulos, G., Davrazos, G., & Kotsiantis, S. (2024). Explainable Artificial Intelligence-Based Decision Support Systems: A Recent Review. Electronics, 13(14), 2842.

[17] Bayer, S., Gimpel, H., & Markgraf, M. (2022). The role of domain expertise in trusting and following explainable AI decision support systems. Journal of Decision Systems, 32(1), 110-138.

[18] Zhang, S. (2024). Consumer attitudes towards AI-based financial advice: insights for decision support systems (DSS) and technology integration. Journal of Internet Services and Information Security, 14(4), 1-20.

[19] Naiseh, M., Al-Thani, D., Jiang, N., & Ali, R. (2023). How the different explanation classes impact trust calibration: The case of clinical decision support systems. International Journal of Human-Computer Studies, 169, 102941.

[20] Praveen, S. P., Nakka, R., Chokka, A., Thatha, V. N., Vellela, S. S., & Sirisha, U. (2023). A novel classification approach for grape leaf disease detection based on different attention deep learning techniques. International Journal of Advanced Computer Science and Applications (IJACSA), 14(6), 2023.

[21] Biyyapu, N., Veerapaneni, E. J., Surapaneni, P. P., Vellela, S. S., & Vatambeti, R. (2024). Designing a modified feature aggregation model with hybrid sampling techniques for network intrusion detection. Cluster Computing, 27(5), 5913-5931.

Authorized licensed use limited to: Sardar Patel Institute of Technology. Downloaded on January 28,2026 at 04:57:19 UTC from IEEE Xplore. Restrictions apply.